

| Category                             | n  | Mean (pre) | Mean (post) | Mean Diff | W    |
|--------------------------------------|----|------------|-------------|-----------|------|
| Computing Background                 |    |            |             |           |      |
| experience-computer                  | 13 | 5.77       | 6.15        | 0.38      | 3    |
| experience-programming               | 13 | 4.15       | 4.54        | 0.38      | 6    |
| experience-peers                     | 13 | 3.77       | 4.54        | 0.77      | 3.5  |
| Programming Concepts                 |    |            |             |           |      |
| Data/Variable Types                  | 13 | 4.15       | 5.23        | 1.08      | 7    |
| Arithmetic Expressions and Operators | 13 | 4.31       | 5.23        | 0.92      | 7    |
| Branching                            | 13 | 4.38       | 5.08        | 0.69      | 6    |
| Loops                                | 13 | 4.23       | 4.77        | 0.54      | 14   |
| Calling functions/methods            | 13 | 4.31       | 4.92        | 0.62      | 6    |
| Arrays                               | 13 | 3.46       | 4.69        | 1.23      | 7    |
| Writing static functions             | 13 | 3.38       | 4.85        | 1.46      | 3    |
| Writing methods                      | 13 | 3.92       | 5           | 1.08      | 7    |
| Writing classes                      | 13 | 3.46       | 5           | 1.54      | 3    |
| Creating objects                     | 13 | 3.54       | 4.85        | 1.31      | 5    |
| PDC Concepts                         |    |            |             |           |      |
| parallel                             | 13 | 2.08       | 3.62        | 1.54      | 3.5  |
| distributed                          | 13 | 2          | 3.08        | 1.08      | 3    |
| event                                | 13 | 3.08       | 3.77        | 0.69      | 16.5 |
| Computing Attitudes                  |    |            |             |           |      |
| computing interest                   | 13 | 6.15       | 6.31        | 0.15      | 6    |
| PDC interest                         | 13 | 5.77       | 5.85        | 0.08      | 16.5 |
| PDC importance                       | 13 | 5.85       | 6.15        | 0.31      | 5.5  |
| computing understanding              | 13 | 4.77       | 5.31        | 0.54      | 3.5  |
| PDC understanding                    | 13 | 3.46       | 4.08        | 0.62      | 13   |

n = 13 records processed

| p-value | Sig?  | Cliff delta | interpretation |
|---------|-------|-------------|----------------|
| 0.188   | FALSE | -0.17       | small          |
| 0.25    | FALSE | -0.15       | small          |
| 0.027   | TRUE  | -0.26       | small          |
| 0.021   | TRUE  | -0.25       | small          |
| 0.043   | TRUE  | -0.19       | small          |
| 0.133   | FALSE | -0.14       | negligible     |
| 0.363   | FALSE | -0.08       | negligible     |
| 0.125   | FALSE | -0.14       | negligible     |
| 0.043   | TRUE  | -0.3        | small          |
| 0.006   | TRUE  | -0.44       | medium         |
| 0.045   | TRUE  | -0.26       | small          |
| 0.012   | TRUE  | -0.41       | medium         |
| 0.047   | TRUE  | -0.33       | small          |
| 0.003   | TRUE  | -0.59       | large          |
| 0.023   | TRUE  | -0.44       | medium         |
| 0.169   | FALSE | -0.23       | small          |
| 0.531   | FALSE | -0.24       | small          |
| 0.961   | FALSE | -0.03       | negligible     |
| 0.406   | FALSE | -0.22       | small          |
| 0.109   | FALSE | -0.17       | small          |
| 0.289   | FALSE | -0.23       | small          |